

- 10** The half-life of a radioactive isotope **P** is twice that of another radioactive isotope **Q**. In a sample of rock, it is found that the number of atoms of **Q** is 4 times that of **P**.

What will be the ratio of the number of atoms of **P** to the number of atoms of **Q** in the rock after two half-lives of **P**?

A $\frac{1}{16}$

B $\frac{1}{4}$

C $\frac{1}{2}$

D 1